minPoints (M for short, in the formulas below), the minimum number of points needed to define a dense region, these definitions also depend on those hyperparameters.

$$\texttt{core-distance}_{\varepsilon,M}(p) = \begin{cases} Undefined & \texttt{if } |N_\varepsilon(p)| < M \\ M\texttt{-th smallest distance in } |N_\varepsilon(p)| & \texttt{otherwise} \end{cases}$$

Naturally, if a point is not a core point (and so less than M points are in its ε-neighborhood), its core distance is undefined. Conversely, its core distance will be the distance of its M-th nearest neighbor[21]; if so happens, core distance can be interpreted as the minimum value that we can assign to the parameter ε for p to be a core point (given a fixed value for minPoints).

Once we have defined the core distance, we can use it to define a new kind of metric, the reachability distance between two points:

$$\texttt{reachability-distance}_{\varepsilon,M}(p,q) = \begin{cases} Undefined & \texttt{if } |N_\varepsilon(q)| < M \\ max(\texttt{core-distance}_{\varepsilon,M}(q)), distance(q,p)) & \texttt{otherwise} \end{cases}$$

Reachability distance is not symmetric: given a point q, the reachability distance of p from q can be interpreted as the minimum value of ε for which p is density-reachable from q; therefore, it must be at least p's core distance (for the closest minPoints to p), or the actual distance between p and q; moreover, if p and s are nearest neighbors (in particular if there is no point closer to p than q), then this value is the smallest value we can assign to ε in order for p and q to belong to the same cluster.